Q1 – 6: Converse statement

Q1 – 6: Prove the following and state without proof if the converse statement is true.

Q1. When $x=\sqrt{2}$, $\dfrac{2+x}{2-x}=3+2\sqrt{2}$.

Proof. Sub $x=\sqrt{2}$ into $\dfrac{2+x}{2-x}=\dfrac{2+\sqrt{2}}{2-\sqrt{2}}=\dfrac{2+\sqrt{2}}{2-\sqrt{2}}\times\dfrac{2+\sqrt{2}}{2+\sqrt{2}}=\dfrac{6+2\sqrt{2}}{4-2}=3+2\sqrt{2}$.

The converse statement is true:

$$\frac{2+x}{2-x}=3+2\sqrt{2}\Rightarrow\frac{2+x-(3+2\sqrt{2})(2-x)}{2-x}=0$$

$$\Rightarrow\frac{-4(1+\sqrt{2})+2(2+\sqrt{2})x}{2-x}=0\Rightarrow x=\frac{2(1+\sqrt{2})}{2+\sqrt{2}}=\sqrt{2},x\neq 2$$

Q2. Given ΔABC such that $AB=c=a=BC$. Then $\measuredangle A=\measuredangle C$.

Proof. From the sine laws, $\dfrac{\sin A}{a}=\dfrac{\sin C}{c}\Rightarrow\sin A=\sin C$ if $a=c$. Hence,

$$\sin A=\sin C\Rightarrow\measuredangle A=\measuredangle C$$

(because ABC is a triangle and their angle is less than or equal 90 degree).

The converse statement is true:

$$\measuredangle A=\measuredangle C\Rightarrow\sin A=\sin C\Rightarrow\frac{a}{\sin A}=\frac{c}{\sin C}\Rightarrow a=c.$$

Q3. If $y=f(x)$ has a maximum point at $x=a$, then $\left.\dfrac{dy}{dx}\right|_{x=a}=0$.

Intuitive proof. At a maximum/minimum point, the slope is zero (the tangent line is horizontal). Hence, the gradient is zero at the point, so $\left.\dfrac{dy}{dx}\right|_{x=a}=0$.

The converse statement is not true, because a minimum point also has the same property (and also inflection point may also have the same property).

Q4. Let $\dfrac{1}{u}+\dfrac{1}{v}=\dfrac{1}{f}$. If $u\to\infty$ then $f\to v$.

Proof. If $u\to\infty$ and hence $\dfrac{1}{f}\to\dfrac{1}{v}+0$, or $f\to v$.

Converse is not true, because $u\to-\infty$ also implies that $f\to v$.

Q5 – 15: Prove by induction.

Q5. $\sum_{r=1}^{n} r = \frac{n}{2}(n+1)$.

Proof. When $n = 1$, LHS $= 1 =$ RHS.

Assume the result is true for $n = k > 1$, so we have $\sum_{r=1}^{k} r = \frac{k}{2}(k+1)$.

When $n = k+1$,

$$\sum_{r=1}^{k+1} r = \sum_{r=1}^{k} r + (k+1) = \frac{k}{2}(k+1) + (k+1) = \left(\frac{k}{2}+1\right)(k+1) = \frac{1}{2}(k+1)(k+2).$$

Thus, the result is true for all positive integer n.

Q6. 6 divides $n^3 - n$ for all positive integer n.

Proof. When $n = 1$, $6 \mid 0$.

Assume that result is true for $n = k > 1$, so we have $6 \mid k^3 - k$.

Now, when $n = k+1$, we have

$$(k+1)^3 - (k+1) = k^3 - k + 3k(k+1).$$

Here, $k^3 - k$ is a multiple of 6 and since $k(k+1)$ is even (because k and $k+1$ are consecutive integers), it follows that $3k(k+1)$ is a multiple of 6. Thus, the result is true for all positive integer n.

Q7. $\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}$

Proof. When $n = 1$, LHS $= \frac{1}{2} =$ RHS.

Assume that the result is true for $n = k > 1$, i.e., $\sum_{r=1}^{k} \frac{1}{r(r+1)} = \frac{k}{k+1}$.

When $n = k+1$, we have

$$\sum_{r=1}^{k+1} \frac{1}{r(r+1)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{k+1}{k+2}.$$

Thus, the result is true for all positive integer n.

Q8. $7 \mid 2^{n+2} + 3^{2n+1}$ for all positive n.

Proof. When $n = 1$, $7 \mid 2^3 + 3^3 = 35$.

Assume that the result is true for $n = k > 1$, i.e., $7 \mid 2^{k+2} + 3^{2k+1}$.

When $n = k + 1$, we have

$$2^{k+3} + 3^{2k+3} = 8(2^k) + 27(3^{2k}) = 2(2^{k+2} + 3^{2k+1}) + 21(3^{2k})$$

is a multiple of 7 because $7 \mid 2^{k+2} + 3^{2k+1}$ and $7 \mid 21(3^{2k})$. Thus, the result is true for all positive integer n.

Q9. $\dfrac{d^n}{d\theta^n}\Big(\sin a\theta\Big) = a^n \sin\left(a\theta + \dfrac{n\pi}{2}\right)$ for all positive integer n.

Proof. Note that $\cos\theta = \sin\Big(\theta + \frac{\pi}{2}\Big)$.

When $n = 1$, $\dfrac{d}{d\theta}\Big(\sin a\theta\Big) = a\cos a\theta = a\sin\left(a\theta + \dfrac{\pi}{2}\right)$. So, the result is true.

In fact, when $n = 0$, $\dfrac{d^0}{d\theta^0}\Big(\sin a\theta\Big) = a^0 \sin\Big(a\theta\Big)$ is also true.

Assume that the result is true for $n = k > 1$. Now when $n = k + 1$, we have

$$\frac{d^{k+1}}{d\theta^{k+1}}\Big(\sin a\theta\Big) = \frac{d}{d\theta}\left[a^k \sin\left(a\theta + \frac{k\pi}{2}\right)\right] = a^{k+1}\cos\left(a\theta + \frac{k\pi}{2}\right) = a^{k+1}\sin\left(a\theta + \frac{(k+1)\pi}{2}\right).$$

Thus, the result is true for all positive (non-negative) integer n.

Q10. $2^n > 2n$ for all positive integer more than 2.

Proof. When $n = 3$, $2^3 = 8 > 2(3) = 6$. Assume that inequality is true for $n = k > 2$, i.e., $2^k > 2k$. Then when $n = k + 1$,

$$\begin{aligned} & k > 2 \\ \Rightarrow\ & 2k > k > 2 \\ \Rightarrow\ & 2k + 2k > 2 + 2k \\ \Rightarrow\ & 2^{k+1} = 2(2^k) > 2(2k) > 2(k+1) \quad \because 2^k > 2k \end{aligned}$$

Thus, the result is true for all positive (non-negative) integer n.

Q11. $u_{r+1} = \frac{2}{3}u_r - \frac{1}{3} \ \ \forall r \in \mathbb{Z}^+$ where $u_1 = 0$. Then $u_n = 3\left(\frac{2}{3}\right)^n - 1 \ \ \forall n \in \mathbb{Z}^+$.

When $r = 2$, $u_2 = \frac{1}{3} = 3\left(\frac{2}{3}\right)^2 - 1$. So, the result is true.

Suppose the result is true for $r = k$ for some positive integer $k > 1$.

Now when $r = k + 1$,

$$u_{k+1} = \frac{2}{3}u_k - \frac{1}{3} = \frac{2}{3}\left(3\left(\frac{2}{3}\right)^k - 1\right) - \frac{1}{3} = 2\left(\frac{2}{3}\right)^k - 1 = 3\left(\frac{2}{3}\right)^{k+1} - 1.$$

Thus, the result is true for all positive integer n.

Q12. $\int x^n dx = \dfrac{x^{n+1}}{n+1} + C$ for all positive integer n, C is a constant.

Proof. $\int x dx = \dfrac{x^2}{2} + C$ is true for $n = 1$.

Assume that the result is true for $n = k$, i.e., $\int x^k dx = \dfrac{x^{k+1}}{k+1} + C$.

Now when $n = k + 1$, we have

$$\begin{aligned}
&\int x^{k+1} dx && \Rightarrow (1 + \tfrac{1}{k+1})\int x^{k+1} dx = \frac{x^{k+2}}{k+1} + D \\
&= \int x \cdot x^k dx && \Rightarrow \tfrac{k+2}{k+1}\int x^{k+1} dx = \frac{x^{k+2}}{k+1} + D \\
&= x\left(\tfrac{x^{k+1}}{k+1} + C\right) - \int 1 \cdot \left(\tfrac{x^{k+1}}{k+1} + C\right) dx && \Rightarrow \int x^{k+1} dx = \frac{x^{k+2}}{k+2} + \frac{D}{k+2} \\
&= \tfrac{x^{k+2}}{k+1} + Cx - \tfrac{1}{k+1}\int x^{k+1} dx - Cx + D && \Rightarrow \int x^{k+1} dx = \frac{x^{k+2}}{k+2} + K, K = \text{constant}
\end{aligned}$$

Thus, the result is true for all positive integer n.

Q13. Prove by (a) deduction, (b) induction, that

$$\sum_{r=1}^{n} ap^r = \frac{a(1-p^n)}{1-p} \forall n \in \mathbb{Z}^+.$$

Proof. Since $\sum_{r=1}^{n} ap^r = a\sum_{r=1}^{n} p^r$, it follows that, without loss of its generality, we prove that $\sum_{r=0}^{n} p^r = \dfrac{1-p^{n+1}}{1-p} \forall n \in \mathbb{Z}^+$.

(a) When $n = 1$, $\sum_{r=0}^{1} p^r = 1 + p = \dfrac{1-p^2}{1-p}$.

When $n = 2$, $\sum_{r=0}^{2} p^r = 1 + p + p^2 = \dfrac{1-p^3}{1-p}$.

When $n=3$, $\sum_{r=0}^{3} p^r = 1+p+p^2+p^3 = \dfrac{1-p^4}{1-p}$.

Hence, by deduction, $\sum_{r=0}^{n} p^r = \dfrac{1-p^n}{1-p} \forall n \in \mathbb{Z}^+$.

(b) When $n=1$, LHS $= \sum_{r=0}^{1} p^r = 1+p = \dfrac{1-p^2}{1-p} =$ RHS.

Suppose the result is true for $n=k$, i.e., $\sum_{r=0}^{k} p^r = \dfrac{1-p^{k+1}}{1-p}$ for some positive integer k.

Then when $n=k+1$, we have $\sum_{r=0}^{k+1} p^r = \dfrac{1-p^{k+1}}{1-p} + p^{k+1} = \dfrac{1-p^{k+2}}{1-p}$. Thus, the result is true for all positive integer n.

Q14. $\dfrac{d}{dx}\left(x^{-n}\right) = -nx^{-n-1}$ for all positive integer n.

Proof. When $n=1$, the result is true: $\dfrac{d}{dx}\left(x^{-1}\right) = -x^{-2}$.

Suppose the result is true for $n=k$, where k is a positive integer >1.

When $n=k+1$, we have

$$\frac{d}{dx}\left(x^{-k-1}\right) = \frac{d}{dx}\left(x^{-k}\cdot x^{-1}\right) = -kx^{-k-1}\cdot x^{-1} + (-x^{-2})\cdot x^{-k} = -(k+1)x^{-k-2}.$$

Thus, the result is true for all positive integer n.

Q15. If $A = \begin{pmatrix} 2 & a \\ 0 & 1 \end{pmatrix}$, $a \in \mathbb{R}$, then $A^n = \begin{pmatrix} 2^n & (2^n-1)a \\ 0 & 1 \end{pmatrix}$ for every positive integer n.

Proof. When $n=1$, then $A = \begin{pmatrix} 2 & (2^1-1)a \\ 0 & 1 \end{pmatrix}$. The result is true.

Suppose the result is true for $n=k$, where k is a positive integer >1. Then when $n=k+1$, we have

$$A^{k+1} = A^k A = \begin{pmatrix} 2^k & (2^k-1)a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2^{k+1} & (2^{k+1}-1)a \\ 0 & 1 \end{pmatrix}.$$

Thus, the result is true for all positive integer n.

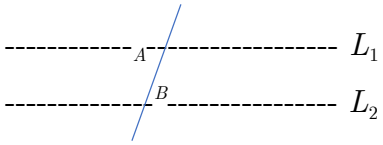
Q1 – 5: Proof by contrapositve

Q1 – 5: Prove the following by contrapositive.

Q1. If n is a natural number such that n^2 is odd (perfect square), then n is odd.

Proof. Suppose n is even, then $n = 2k$ for some positive integer k. Then $n^2 = 4k^2$ is even.

Q2. Let L_1 and L_2 be two non-overlapping lines. If $\measuredangle A = \measuredangle B$, then L_1 and L_2 do not intersect.



Proof. Suppose L_1 and L_2 intersects at one point. Then the two lines are not parallel. Therefore, $\measuredangle A \neq \measuredangle B$.

Q3. If n is a perfect number, then n is not a prime number.

Proof. Suppose n is prime. Then the positive factors of n are 1 and n. Obviously $n \neq 1$ and so n is not a perfect number.

Q4. If two lines are perpendicular, the product of their gradients is –1.

Proof. Let L_1 and L_2 be two lines with gradients $m_1 = \tan\theta_1$ and $m_2 = \tan\theta_2$, respectively, where θ_1, θ_2 are angles of L_1 and L_2, measured from the x-axis, Ox. Suppose $m_1 m_2 \neq -1$. Then $\tan\theta_1 \tan\theta_2 \neq -1$. Now

$$\tan\left(\theta_1 - \theta_2\right) = \frac{\tan\theta_1 - \tan\theta_2}{1 + \tan\theta_1 \tan\theta_2} \neq \frac{\tan\theta_1 - \tan\theta_2}{0}.$$

Hence, $\theta_1 - \theta_2 \neq \dfrac{\pi}{2}$. Thus, L_1 and L_2 are not perpendicular.

Q5. If $(x-2)(1-x) > 0$ then $1 < x < 2$.

Proof. Suppose $x \leq 1$ or $x \geq 2$. We consider the following two cases:

Case 1. $x \leq 1$: Then $x - 2 \leq 1 - 2 = -1 < 0$ and $1 - x \geq 0$. Thus, $(x-2)(1-x) \leq 0$.

Case 2. $x \geq 2$: Then $x - 2 \geq 0$ and $1 - x \leq 1 - 2 = -1 < 0$. Thus, $(x-2)(1-x) \leq 0$.

Q6 – 8: Prove by contradiction.

Q6. $\sqrt{3}$ is irrational.

Proof. First, we prove the fact $3 \mid a^2 \Rightarrow 3 \mid a$, where a is an integer.

Let $a = 3n+1$ or $a = 3n+2$. Then

$$a^2 = 9n^2 + 6n + 1 = 3(3n^2 + 2n) + 1 \text{ or}$$

$$a^2 = 9n^2 + 12n + 4 = 3(3n^2 + 4n + 1) + 1.$$

Hence, 3 does not divide a^2.

Suppose $\sqrt{3} = \frac{m}{n} \in \mathbb{Q}$ for some $m, n \in \mathbb{Z} \setminus \{0\}$. Assume that $\gcd(m,n) = 1$. Then

$$3n^2 = m^2 \Rightarrow 3 \mid m^2 \Rightarrow 3 \mid m \Rightarrow m = 3k \text{ for some } k \in \mathbb{Z}.$$

Substitute this to get

$$3n^2 = 9k^2 \Rightarrow n^2 = 3k^2 \Rightarrow 3 \mid n^2 \Rightarrow 3 \mid n \Rightarrow n = 3j \text{ for some } j \in \mathbb{Z}.$$

This is a contradiction because $\gcd(m,n) \geq 3 \neq 1$.

In general, $p \mid a^2 \Rightarrow p \mid a$ where p is a prime. To prove this, let

$$a \equiv k \pmod{p},\ k = 1, 2, \ldots, p-1.$$

$$\text{Then } a \equiv k^2 \not\equiv 0 \pmod{p},\ k = 1, 2, \ldots, p-1.$$

Suppose $\sqrt{p} = \frac{m}{n} \in \mathbb{Q}$ for some $m, n \in \mathbb{Z} \setminus \{0\}$. Assume that $\gcd(m,n) = 1$. Then

$$pn^2 = m^2 \Rightarrow p \mid m^2 \Rightarrow p \mid m \Rightarrow m = pk \text{ for some } k \in \mathbb{Z}.$$

Substitute this to get

$$pn^2 = p^2k^2 \Rightarrow n^2 = pk^2 \Rightarrow p \mid n^2 \Rightarrow p \mid n \Rightarrow n = pj \text{ for some } j \in \mathbb{Z}.$$

This is a contradiction because $\gcd(m,n) \geq p > 1$.

Q7. If $x^2 - 3x + 2 < 0$ then $1 < x < 2$.

Proof. Note that $\sim\big(p \rightarrow q\big) \equiv \ \sim\big(\sim p \vee q\big) \equiv p \wedge \sim q$. So we assume that

$$x^2 - 3x + 2 = (x-1)(x-2) < 0\text{; and negation of } 1 < x < 2\text{; which means}$$

$$(1-x)(x-2) > 0\text{; and } x \leq 1 \text{ or } x \geq 2.$$

Case 1. $x \leq 1$: Then $x - 2 \leq 1 - 2 = -1 < 0$ and $1 - x \geq 0$. Thus, $(x-2)(1-x) \leq 0$.

Case 2. $x \geq 2$: Then $x - 2 \geq 0$ and $1 - x \leq 1 - 2 = -1 < 0$. Thus, $(x-2)(1-x) \leq 0$.

Both cases arrive at a contradiction. Hence, the proof is completed.

Q8. For all $x > 0$, $x + \dfrac{1}{x} \geq 2$.

Proof. Let k be real where $0 < k < 2$. Suppose $\exists x \in \mathbb{R}^+$ such that

$$x + \frac{1}{x} = k \Rightarrow x^2 - kx + 1 = 0\,.$$

Now $k^2 - 4 \geq 0$ (real solutions: $b^2 - 4ac \geq 0$), contradicts to $0 < k < 2$.

Q9 – 15: Disprove by a counter example.

Q9. For any two real numbers a and b, $a \div b = b \div a$.

Disproof. Take $a = 1 = b - 1$. Then $a \div b = \frac{1}{2} \neq 2 = b \div a$.

Q10. For any two vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}$.

Disproof. Take $\mathbf{a} = (1\,0\,0), \mathbf{b} = (0\,1\,0)$, then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = \mathbf{b} \times \mathbf{a}\,.$$

Note: In general, $\mathbf{a} \times \mathbf{b} = -\big(\mathbf{b} \times \mathbf{a}\big)$ is true.

Q11. For all real values of a and b, $a - b > 0 \Rightarrow a^2 - b^2 > 0$.

Disproof. Take $a = 2 = b + 5$, then $a - b = 5 > 0$ but $a^2 - b^2 = 4 - 9 = -4 \not> 0$.

Note that it is sufficient to disprove by choosing a counter example where $a - b > 0$ but $a^2 - b^2 = 0$. For example, take $a = 1 = -b$.

Q12. Let $ax^2 + bx + c = 0$ be an equation, where a, b and c are real. Then the roots of the equation are real.

Disproof. Take $a = 1 = b = c$. Then $b^2 - 4ac = -3 < 0$, which implies that roots of the equation are not real.

Q13. For any two vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a} \times \mathbf{b} = \mathbf{0} \Rightarrow \mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$.

Disproof. Take $\mathbf{a} = (1\,1\,0) = \mathbf{b}$. Then $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \neq \mathbf{0}$.

Q14. If $f''(x=a)=0$, then $f(x)$ has a point of inflexion when $x=a$.

Disproof. Take $f(x)=x^4$. Then $f''(0)=0$ but $x=0$ is not an inflexion point but a minimum point.

Q15. Let A, B and C be three matrices that are compatible for the products AB and AC. If $AB=AC$ then $A=0$ or $B-C=0$.

Disproof. Let $A=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}, B=\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}, C=\begin{pmatrix}0 & 0\\ 0 & 2\end{pmatrix}$. Then $AB=AC$ but $A\neq 0$ and $B\neq C$.

Q16. Give counter example to show that the following statements are false.

(a) The sum of two unequal, positive, irrational numbers is irrational.

(b) The product of two unequal, positive, irrational numbers is irrational.

Disproof. (a) Take $x=\sqrt{2}, y=2-\sqrt{2}\in\mathbb{Q}'$. Then $x+y=2\in\mathbb{Q}$.

(b) Take $x=\sqrt{2}, y=2\sqrt{2}\in\mathbb{Q}'$. Then $xy=4\in\mathbb{Q}$.